

A. Increase or Smash

time limit per test: 1 second
 memory limit per test: 1024 megabytes

Geumjae has an array a consisting of n zeros. His goal is to transform it into a given target array using a minimum number of operations.

He can perform the following two types of operations any number of times, in any order:

- Increase:** Choose any positive integer x . Increase *all* elements of the array a by x . In other words, he chooses a positive integer x , and for each i ($1 \leq i \leq n$), he replaces a_i with $a_i + x$.
- Smash:** Set *some* elements (possibly none or all) of the array a to 0. In other words, for each i ($1 \leq i \leq n$), he either replaces a_i with 0 or leaves it as before.

Given the final target state of the array a , find the minimum total number of operations (both **Increase** and **Smash**) Geumjae needs to perform.

It can be shown that for any given final array, a sequence of operations always exists.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \leq t \leq 1000$). The description of the test cases follows.

The first line contains a single integer n ($1 \leq n \leq 100$) — the number of elements in the array a .

The second line contains n integers $a_1, a_2, \dots, a_n$ ($1 \leq a_i \leq 100$) — the elements of the target array a .

Output

For each test case, output a single integer — the minimum number of operations required.

Example

input	Copy
3 3 1 1 3 1 100 9 9 9 3 2 4 4 8 5 3	
output	Copy
3 1 11	

Note

Explanation of the first test case:

The target array is $[1, 1, 3]$. A possible sequence of 3 operations (which is the minimum) is:

- Initially, the array is $[0, 0, 0]$. After an **Increase** operation with $x = 2$, the array becomes $[2, 2, 2]$.
- Next, after a **Smash** operation on the first two elements, the array becomes $[0, 0, 2]$.
- Finally, after an **Increase** operation with $x = 1$, the array becomes $[1, 1, 3]$.

We used 2 **Increase** operations and 1 **Smash** operation for a total of 3 operations.

Squarepoint Challenge (Codeforces Round 1055, Div. 1 + Div. 2)

Contest is running

02:14:04

Contestant



→ Submit?

Language: Python 3.13.2
 Almost always, if you send a solution on PyPy, it works much faster

Choose file: Choose File No file chosen

Be careful: there is 50 points penalty for submission which fails the pretests or resubmission (except failure on the first test, denial of judgement or similar verdicts). "Passed pretests" submission verdict doesn't guarantee that the solution is absolutely correct and it will pass system tests.

Submit

→ Score table

	Score
Problem A	500
Problem B	1500
Problem C	1500
Problem D	2500
Problem E	3000
Problem F	4000
Problem G	4000
Problem H1	4000
Problem H2	2500
Successful hack	100
Unsuccessful hack	-50
Unsuccessful submission	-50
Resubmission	-50

* If you solve problem on 00:00 from the first attempt

Explanation of the second test case:

The target array is [100]. A single **Increase** operation with $x = 100$ gives the target array.

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